

Predictive Modeling of Monthly Electricity Consumption using Big Data Analytics

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Github Link: <https://github.com/Dooboofloo/CS6304-Big-Data-Project.git>

1 Abstract

Creating accurate predictions for electricity consumption is essential for effective energy management, especially with increasing energy demands due to new technologies. The goal of this project is to develop a predictive model for long-term monthly electricity consumption using Big Data Analytics and Temporal Convolutional Networks that outperforms machine learning, deep learning, and other TCN approaches that have been tested in previous research. Although previous research has focused on short-term electricity consumption, our model will be designed to focus on less explored long-term electricity consumption. We use TCNs to account for external factors such as weather and holidays, and to test multiple approaches including sliding window segmentation to reduce overfitting and normalization to help with model generalization. Our final model achieved an RMSE of 2,9112.27 MW, MAE of 2,252.37 MW, and WAPE of 8.83%. These results demonstrate a strong predictive accuracy for long-term electricity forecasting. This reinforces our stance that the TCN-based approach provides a robust foundation for strategic energy planning and extended forecasting of energy needs in long-term timeframes.

2 Introduction

In our growing society as technology advances the world's electricity consumption continues to shift. With the implementation of technology such as smart meters, monitoring systems, and IoT devices, large amounts of electrical usage data is being created. From this data comes the opportunity to apply new methods of data analytics to catch trends within the data and improve our ability to forecast electrical needs. Accurate forecasting results in reduced operational costs, ensures reliable service for consumers, and maintains a balance between electrical supply and consumption.

Recent work has handled this issue from the usage of machine learning and deep learning methods. Wang et al. [7] used these methods to capture electrical data on a city-scale shown to be better performing than statistical approaches. Similarly, Lee et al. [2] also carried out several machine learning methods over electrical data and came to the conclusion of neural network methods being a more optimal solution for prediction than regression techniques. Matos et al. [3] also carried out a machine learning-based system for forecasting showing real-world application potential.

What our work aims to accomplish is to highlight the recent work done on this problem using Temporal Convolutional Networks (TCNs) as recent work shows their large benefit. Lara-Benítez et al. [1] displayed TCN's potential to model long-term temporal dependencies with parallel optimal efficiency, outperforming other processes like LSTMs. Shaikh et al. [5] utilized TCN's ability to take in extraneous factors to make a prediction method that takes into account seasons and how it affects consumption. These strengths validate why TCNs are a fitting choice for improving monthly consumption forecasting.

This project aims to build upon the findings relative to TCN in these papers by developing a model that can predict monthly electricity usage based on consumption data. The goal is to improve forecasting accuracy while enabling a more dynamic management of electrical resources. When predictions are reliable suppliers are more prepared to anticipate demand and reduce strain on the overall power system, thus supporting smarter energy distribution.

3 Problem Statement

The primary objective of this project is to produce a model which effectively predicts monthly electricity consumption using big data analytics. By anticipating energy consumption, the model should inform the user about upcoming load usage, allowing for better preparation and resource allocation. We consider various external factors as input variables to our model to more accurately predict power consumption relative to weather cycles and population.

4 Discussion of Papers

The work done in “Temporal Convolutional Networks Applied to Energy-Related Time Series Forecasting” [1] demonstrates the utility of TCN’s and why they are a good basis for forecasting energy-related time series. The work done highlights TCN’s causal convolution enabling long-range dependencies to be modeled efficiently. Their findings after applying a TCN to two datasets based on renewable energy and electricity load showed a superior performance for both accuracy and efficiency. Their findings validated that TCNs can successfully capture temporal patterns that span weeks or months, which lends itself to the measurements of time we are wanting to take with our model. TCNs were also able to allow exogenous features such as holidays or weather to be factored in the data and shown in the results. The results from their

work indicated that TCNs had a lower Mean Absolute Error (MAPE) and Root Mean Square Error (RMSE) than LSTMs with faster prediction times, but took longer to work through the training process. A core setback for this paper was the limitation of using only two datasets from a single country and the lack of comparing the TCN against non-neural approaches in order to validate a significant improvement over baseline methods such as gradient boosting.

Other work done in TCNs does possess this comparative element that validates TCNs over other models. “A new approach to seasonal energy consumption forecasting using temporal convolutional networks” proposes a TCN approach that compares it against a Long Short-Term Memory network (LSTM), a variety of neural networks [5]. Traditional neural networks like LSTM suffer memory retention issues when modeling long-temporal dependencies and TCN’s ability to process inputs in parallel, reduces this inference time. Their findings saw that the TCN performed better based on mean absolute deviation and symmetric mean absolute percentage error, outperforming LSTM. This paper benefits our research in that it deals with seasonal energy consumption forecasting and again supports why TCN is a justifiable choice compared to other recurrent neural networks.

Traditional machine-learning methods have also been applied to forecasting electrical consumption. The work done in “A Comparative Study of Forecasting Electricity Consumption Using Machine Learning Models” examines various non-deep learning models for forecasting [2]. Methods such as Artificial Neural Networks, Adaptive Neuro-Fuzzy Inference System (ANFIS), Least Squares Support Vector Machines (LSSVM), and Fuzzy Time Series (FTS) are applied to the problem. A major benefit to this paper’s work is its utilization of electricity records from a variety of countries giving diversity and authenticity to the data results. The paper judges each method against MAPE and RMSE then correlates it against actual

consumption values. This paper benefits our work by providing a foundation for how our method can compare to traditional ones. It also provides insight into the nuances of each country's consumption patterns, but at the same time having a dataset this diverse results in no one model being the overall best choice. It makes it harder to come to a concise conclusion, when several models do better when applied to different countries' records. A core weakness of this paper is the use of monthly data rather than daily or weekly data which machine-learning models struggle with resulting in limited granularity and detect short-term variations. “A Machine Learning-Based Electricity Consumption Forecast and Management System for Renewable Energy Communities” also applied a machine-learning approach with the utilization of XGBoost in order to incorporate exogenous variables [3]. While the approach performed well for the short-term it suffered in capturing long-term data. TCN solves this issue by being able to be applied to sequences of any length. They can calculate meaningful trends, maintain a strong predictive performance, and factor in exogenous data.

Wang et al. [7] forecasts electricity consumption on a metropolitan scale across three U.S. cities using models such as LightGBM, random forests, and neural networks. Their results showed that LightGBM had the highest accuracy with a coefficient of variation of the root mean square error of 4.1% in New York using exogenous factors in consumption as a reason for this result. This work most prominently displays the effectiveness of traditional and ML models, but similar to the other papers mentioned, does not handle long-range temporal dependencies explicitly. TCNs are well-suited to handle this issue and surpass the limitations of ML approaches offering potentially improved predictive performances.

From these studies, TCNs are shown to have consistent results for time-series forecasting in the field of energy. When combined with other methods to reduce data noise it can be

exceptionally effective. Collectively, these works do support that traditional methods and methods that incorporate deep learning can be the solution to forecasting electricity consumption effectively. Although, TCNs can provide a solution that is better suited to be scaled and provide more accurate long-term predictions. The core motivation for our work is the gap in knowledge for long-term forecasting with TCNs and the incorporation of external factors, sliding window segmentation, and normalization into the method to predict consumption across varied datasets.

5 Comparison Problem Statement

The goal of our work is to use a Temporal Convolutional Network to forecast long-term monthly electricity consumption. So far related research has dealt specifically with short-term energy consumption, we believe that by applying improvements to forecasting methods in a more long-term format can enable us to find more improvements in the field. These improvements can also offer a significant advantage to industry as energy demands are constantly growing year after year. Our intent is to thoroughly test our model to understand if we can improve upon the results already observed in related studies. One approach that we plan to test is a sliding window, which we believe will reduce overfitting, by allowing us to train the model on many smaller sections of time, rather than all data at once. We also plan to utilize normalization for the data, allowing us to focus on in what ways electricity use changes over time, rather than by how much it changes.

In order to understand the strength of our model, we compared it against the results of the TCNs that were used in “Temporal Convolutional Networks Applied to Energy-Related Time Series Forecasting” [1].

To test our model’s performance, we implemented a testing framework based on three complementary evaluation metrics: Weighted Average Percentage Error (WAPE), Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). WAPE is intended to account for the magnitude of actual consumption values serving as a comparison across different time periods and consumption levels. By adding MAE and RMSE to WAPE we can ensure cross-study comparability and the typical error magnitude. All tests are set to using inversed-transformed predictions to evaluate performance in the original megawatt scale rather than normalized values.

6 Algorithms and Discussion

The algorithm that we used to solve the long-term prediction problem for Electricity consumption will utilize Temporal Convolutional Network (TCN) architecture for optimized large-scale, long-horizon energy consumption forecasting. The TCN’s parallelizable convolutional layers make it well-suited for high-volume time-series data, enabling distributed training on GPU or multi-node environments, such as S&T’s “Mill”. The dataset that we are applying to our model is a combination of a two datasets to provide our model with all features we want to take into account and mitigate the need for normalization, the components of our dataset are as follows:

- [<https://www.kaggle.com/datasets/albertovidalrod/electricity-consumption-uk-2009-2022>] The dataset contains electrical data ranging from 2009-2024 in the UK. The dataset is 279,000 records. This dataset is pivotal for our long-term monthly focus of our TCN’s forecasting capabilities.[6]
- [<https://www.metoffice.gov.uk/research/climate/maps-and-data/historic-station-data>] The dataset is a public archive of long-term weather data and climate

observations across a number of UK weather stations. This is utilized to offer information about each day such as max temperature, minimum temperature, millimeters of rainfall, or hours of sunlight and then apply them to our TCN as extraneous features. From this dataset we can include environmental features and their influence on energy usage. [4]

These two datasets were combined into a singular dataset csv file with feature labels of the following: year, month, day, hour, nat_usage_mw, temperature max in celsius, temperature minimum in celsius, temperature average in celsius, airfrost_days, rain in millimeters, hours of sunlight, and if it is a holiday.

The pipeline of our model starts at preprocessing which includes ingestion through a distributed framework like PySpark. Data will be enabled to perform parallel loading and transformation by partitioning by region and timestamps. The data partitions will be split chronologically to negate temporal leakage at intervals of 80% for training, 10% for validation, and 10% for testing. Monthly aggregation will be utilized for feature engineering, summon daily readings into monthly usage recaps that store supporting values such as the features applied to our dataset from the weather dataset. Rolling features such as seasonal indexes were also applied to aid trend recognition.

A TCN operates by conducting a hierarchical stack of convolutional layers to build predictions with each layer patterning at different temporal scales. The core principle of TCN is that predictions are based only on observations that came before and never on observations that come after. This quality is insured via causal convolutions that pad the input sequence only on the left side or in other words the past values. When the filter slides across a sequence, it only sees the current and previous values. TCN's are able to use dilated convolution to view previous

data at increasingly larger intervals to capture patterns without having to use an absurd number of layers. An example of how this scales can be explained as a TCN with 4 layers of kernel size 3 and dilations [1,2,4,8] can look back to 61 timestamps. TCN's also have residual connections which let temporal blocks that let the TCN add input directly to output which eliminates the vanishing gradient problem and learn incremental changes over complete changes at each layer.

The TCN architecture for our work is as follows:

- **Architecture Components:**
 - **Input Layer:**
 - Dimensions: 11 features x 720 timesteps in a 15-day hourly window
 - Features:
 - Temporal markers
 - Year
 - Month
 - Day
 - Hour
 - Weather variables:
 - Temperature min/max/avg
 - Air frost days
 - Rainfall
 - Sunlight hours
 - Binary holiday indicator
 - Preprocessing: StandardScalar normalization is applied independently to feature and target variables using the training set statistics only

- **Causal 1-D Convolutions:**

Our TCN Implementation uses causal 1-D convolutions to ensure strict temporal ordering. It is imperative our model predicts based on the past data and not future data when making predictions for a given month.

- Temporal Ordering: maintains strict causality via preventing information leakage from future timestamps
- Implementation: Zero-padding on the left side only for our historical data
- Kernel Size: 3

- **Dilated Convolutions:**

Electricity consumption is strongly affected by seasonal cyclicity such as winter months resulting in more electricity being spent on heating or daylight savings taking effect. Summer causes more air conditioning to be used. Our dilated convolution method aims to capture these annual cycles across the months and years in our dataset.

- Dilatation Schedule: Across four temporal blocks there exists an exponentially increasing pattern of [1,2,4,8]
- Receptive Field: The receptive field spans $(3-1) \times (1+2+4+8)+1=31$ timestamps based on a kernel size of 3 and four temporal layers. This allows the model to observe patterns across more than a full day.
- Multi-scale Learning: Early layers capture hourly patterns, middle catch daily cycles, and deeper layers detect weekly trends

- **Temporal Block Architecture:**

- Dual Convolution Layers: Two consecutive dilated convolutions for hierarchical feature extraction.
- Activation Function: ReLU applied after each convolution to introduce non-linearity.
- Channel Configuration: [64,64,64,64] filters across all blocks, to provide enough capacity for sufficient pattern learning.
- Normalization: Layer normalization applied to stabilize training and ensure consistent features scales, especially for variables like weather which can be vastly different from each other.

○ **Residual Connections:**

Electricity consumption forecasting results in numerous overlapping patterns that must be learned simultaneously. Aggregations using daily, monthly, seasonally, and yearly data all fall into this technical depth. Residual connections allow us to handle these complex interactions without degradation of the gradient.

- Skip Connections: 1x1 convolutions for channel matching when input and output are different, otherwise it uses identity mapping.
- Gradient Flow: Mitigates vanishing gradient problem by enabling direct gradient propagation to earlier layers.
- Additive Learning: Each block can learn residual adjustments in comparison to complete transformations.

○ **Dropout and Layer Normalization:**

To prevent the possibility of overfitting our model based on noise or outliers we used layer normalization and spatial dropout configurations. A dropout rate of 0.3

is applied after each convolutional layer with entire features maps being dropped to preserve spatial correlations. Layer normalization is applied to stabilize training and ensure consistent feature scales especially when features like weather can have many different natural ranges.

- Dropout Rate: 0.3 is applied after each convolutional layer within each temporal block
- Spatial Dropout: whole feature maps are dropped rather than individual neurons in order to preserve spatial correlations in time-series data.
- Early Stopping: 6 epochs of patience with validation loss monitoring to prevent overtraining.

- Output Layer:

- Fully Connected Layer: 64 channels to a single output value via linear projection.
- Temporal Aggregation: Applied specifically to the final timestamp of the sequence.
- Output Scale: A single scalar representing predicted electricity consumption for the hour immediately following the input window.

- **Sliding Window Segmentation:**

- Time series data will be divided into overlapping sequences. For example, a 24-month input may be used for a 1-month prediction. This increases samples while reducing overfitting. For our dataset of months ranging from 2009-2022 we have a 156-month time series. Through using sliding window segmentation we can have multiple overlapping training samples that increases the effective dataset

size while maintaining temporal relationships. It also helps seasonal representation via having a training window span over more than a year allows the model to pick up on seasonal patterns by seeing those patterns more than once in each window.

- **Evaluation Metrics:**

- **Primary:** Weighted Average Percentage Error (WAPE)

- $WAPE = (\sum_t |y_t - \hat{y}_t|) / (\sum_t |y_t|)$

- According to previous reviewed papers, current TCN approaches achieve a 5-15% WAPE for short-term forecasting. A meaningful improvement over baseline statistical methods for us would be a $WAPE < 10\%$. This states that we want our accuracy to be within a 10% margin of the actual value.

- **Secondary:** Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE)

to ensure cross-study comparability

- RMSE: $\sqrt{1/n \sum (y_t - \hat{y}_t)^2}$

- MAE : $1/n \sum |y_t - \hat{y}_t|$

- We consider $RMSE < 3,000 \text{ MW}$ and $MAE < 2,500 \text{ MW}$ to be the success criterion for these values.

- Possessing a RMSE below 3,000 MW indicates that our model avoids large-scale predictive errors where over or under provisioning of electrical resources may take place.
 - Possessing a $MAE < 2,500 \text{ MW}$ tells us the model's average prediction accuracy. Being below 2,500 MW states that the

model's prediction deviates from the actual consumption value by less than 2,500 MW.

In the final stages of training, we will use several optimization techniques, such as using loss functions or early stopping to prevent overfitting to outliers and overfitting on long-sequence data, respectively. Other optimization techniques may be considered as needed, such as normalization or regularization.

The integration of distributed processing and parallelization to our TCN architecture aligns with this course's focus on big-data analytics and parallel computing, as well as highlighting the power of temporal models for real-world forecasting tasks.

7 Proposed Models and Solution

After reviewing potential solutions in past work, which often focused on short-term electricity forecasting, our project was decided to focus on approaches towards long-term forecasting. We aimed to design methods that could apply to spans of months and years, while accounting for the cyclical nature of electricity consumption, since electricity consumption tends to follow similar trends when the same month is considered across multiple years. The following approaches were considered when developing our model:

➤ **Sliding Window / Lag Feature Compensation.**

- Because electricity usage is strongly autocorrelated (that is, it shows a high degree of correlation between successive time windows), we will incorporate “lagged” values into our feature set. For example, consumption values from the previous day, week, month, or even year can be explicitly provided to the network, rather

than relying on the model to develop these inferences on its own. Doing so may influence a bias toward predicting the periodicity inherent to electricity consumption while reducing the risk of overfitting to short-term time windows.

➤ **Normalization by Season.**

- A key flaw of prior work is a lack of contextual handling of seasonal activity, including unusual events such as heatwaves and cold snaps. To address this, we plan on normalizing our dataset with respect to seasonal cycles and weather patterns, likely based on some average seasonal consumption or weather-adjusted baseline (temperature, humidity, etc.).

➤ **Temporal Convolutional Networks (TCNs).**

- Temporal Convolutional Networks offer an efficient alternative to standard models for forecasting, leveraging causal convolutions to capture long-term dependency in the dataset. TCNs have been shown to outperform recurrent networks (those that rely heavily on previous outputs) in tasks involving time-series data. Their ability to model patterns across various time scales makes them particularly well suited for our application, where seasonal cycles may span many timesteps.

➤ **Inclusion of Exogenous Variables**

- To address the lack of context present in prior work, we will include exogenous variables such as temperature, humidity, population, and other variables where available. Neural networks typically excel at integrating auxiliary features with time-series inputs, which can form connections and inferences that often surpass

human reasoning. This should help our model predict variability caused by weather or population behavior rather than simple autocorrelation.

➤ **Regime, Holiday, and Event Modeling**

- One possible approach we could consider in order to improve the robustness of the model is the inclusion of regime indicators that could be used to account for weekdays/weekends, holidays, policy changes, and unexpected notable weather events. Accounting for these events in the model may help to understand outliers in the model, as regime indicators would allow the model to note when abnormal electricity consumption could be expected.

➤ **Model Ensembling**

- One approach that we considered was model ensembling, in which we would combine our TCN model with additional models, such as simpler regression models, to capture different structures in the data. Utilizing model ensembling could help to capture trends not found when only using a TCN model and could improve the robustness of the model.

➤ **Probabilistic and Quantile Forecasting**

- To further improve the robustness of our model, the approach of probabilistic forecasting was considered in order to offer a range of predictions instead of single point predictions for future electricity consumption values. Probabilistic forecasting could be designed for the model in multiple ways, with one approach being quantile forecasting. With quantile forecasts, electricity consumption could be predicted through the use of specific quantiles, such as 10th, 50th, and 90th quantiles. This approach could notably improve the robustness of the model, as it

would improve how well the confidence of the model on specific electricity consumption predictions could be understood.

➤ **Multi-horizon Forecasting**

- Multi-horizon forecasting offers an efficient way to get information about electricity consumption at multiple points in the future with the use of a single prediction. By using this approach, we can predict several future time steps with each input window, such as 1 month, 6 month, and 12 month predictions, making the model more efficient.

8 Experiments and Performance

The experimental aspect of this project was conducted using Missouri S&T's high-performance computing cluster known as "The Mill" which allowed us to have the computational resources to train our TCN model. This enabled our model to possess efficient parallel processing of TCN operations across multiple training epochs. Before creating our model, preprocessing of our data took place where we unified the data from the UK electricity dataset and the UK Met Office historic weather station dataset into one dataset to be pipelined into our model. The merged dataset included 279,000 records with the 11 features mentioned previously in section 6 of this report. Our target variable was national electricity usage measured in megawatts (MW).

The data was partitioned chronologically with 80% going to training our model, 10% for validation, and 10% for testing. The data was split chronologically due to the time-series nature of our data and to test the model on unseen future based data rather than a random datapoint that may have come before or after the training datapoints. The training dataset consisted of around

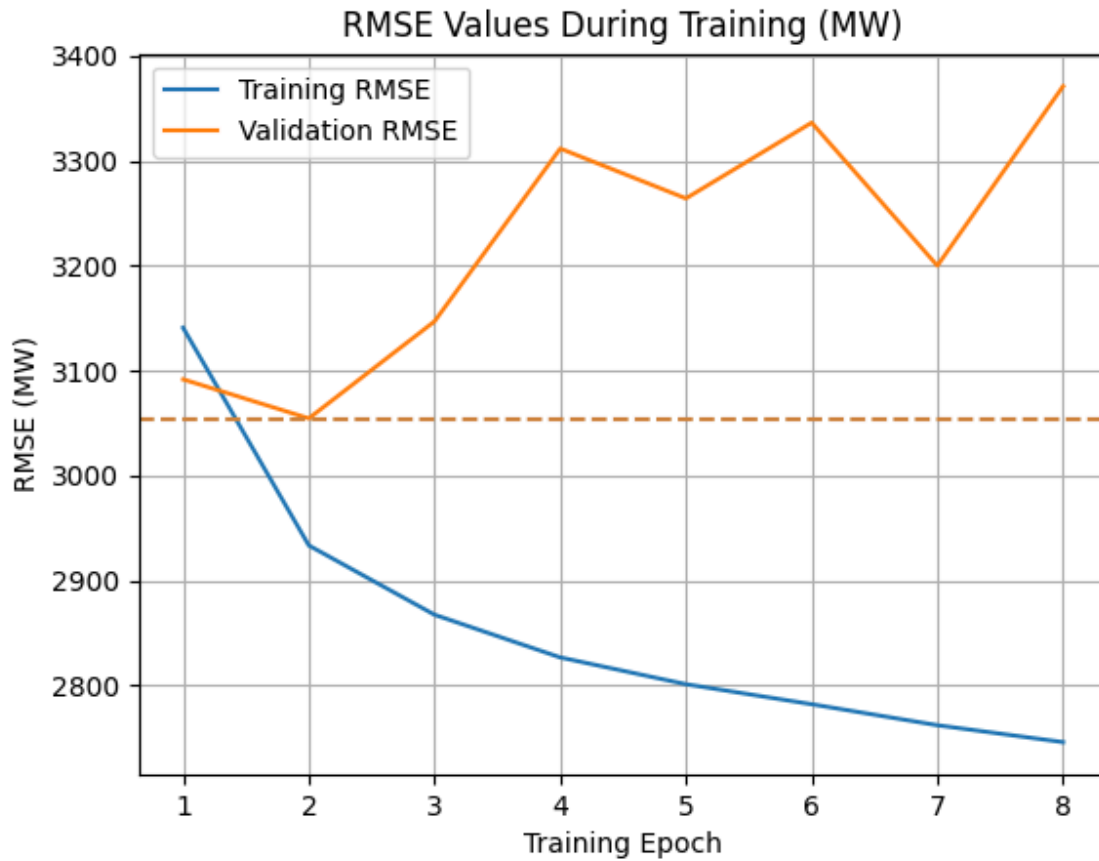
223,200 hourly observations, 27,900 for validation, and 27,900 for testing. Standard Scaler normalization was also applied independently to both feature and target variables. This prevents data leakage from validation and test sets while ensuring that the model learns patterns in normalized space.

The model was trained on the Adam optimizer with a learning rate 1×10^{-3} . Adam was chosen due to its adaptive learning rate capabilities and optimal performance in time-series based data versus other optimizers like SGD. Mean Squared Error (MSE) loss functions were used as the training objective, to penalize larger prediction error metrics in order to prevent massive prediction failures. Training was done in batch sizes of 64 samples which maximised GPU performance while maintaining stable gradient estimates. Overfitting was countered by setting patience at 6 epochs to monitor validation loss. If no improvement is observed in training for 6 consecutive epochs the training is terminated. The model state corresponding to the best validation loss is preserved and restored for final evaluation, thus ensuring that the deployed model represents the optimal generalization performance rather than the final training state.

The best model, which corresponded to the best validation point of epoch 2, was then tested on a test set of 27,900 hourly predictions. Performance of this model gave a result of 2,912.27 MW for RMSE, 2,252.37 MW for MAE, and 8.83% for WAPE. This evaluation was done with regards to the original measurement units. A sample of the prediction values of our model and its final_metrics of the overall performance based on RMSE, MAE, and WAPE can be observed below.

RMSE, MAE, WAPE

2912.2742, 2252.3672, 0.088296674



The above figure shows the RMSE of the model evaluated against the training data and validation data after each epoch of training. The model quickly converged to a minimal validation RMSE of ~3054 MW before rapidly diverging, while the training RMSE continued to decline. This shows clear evidence that, had the model not been implemented with early stopping in mind, it would have surely overfitted to the training data while generalizing.

The value of 8.83% for the WAPE shows that, on average, it has a deviation of less than 9% between forecasted values and actual values. Such a degree of efficiency in long-term forecasting of values further reveals that this model has a consistent performance no matter how different values are. Another reason why it is important for electricity forecasts: it takes into consideration how much the amount of energy being consumed has been, and a 1,000 MW error means different amounts for a 20,000 MW and 30,000 MW usage.

Also, since the RMSE value of 2,912.27 MW is below our set success threshold of 3,000 MW, this ensures that our model does not experience catastrophic failures with regards to prediction. Since RMSE has a quadratic form that gives a lot of emphasis to larger values, it ensures that our model has low chances of making predictions that are grossly inaccurate. This becomes important in a practical application area where such predictions can cause blackouts as well as higher reserve costs.

The value of 2,252.37 MW MAE, which is also below our threshold of 2,500 MW, represents the average absolute error between the predicted and actual values. Unlike in RMSE, in MAE, all values of error are of equal importance. An MAE of 2,252.37 MW on a normal range of 22,000 to 29,000 MW would imply a normal error of 8-10% as derived from the value of WAPE.

9 Executive Summary

Our project involves developing a long-term energy consumption forecasting model using Big Data Analytics and Temporal Convolutional Networks. As global electricity demand continues to increase, especially with increased requirements from new technologies such as

Artificial Intelligence, accurate forecasting has become more useful for managing and allocating energy resources efficiently.

The model that developed is intended to improve upon existing research regarding energy consumption forecasting. Currently, short-term predictions have been explored in more research, and they have generally used machine learning and deep learning techniques to create predictive models. Our model focuses on long-term forecasting, which has been researched less but does provide value for organizations planning how to efficiently manage power in the future.

To prepare data for the model to be trained on, two datasets were used. One set contains UK electrical data from 2009-2024, while the other is a public archive of long-term weather and climate data from multiple UK weather stations. These two datasets were combined into a single csv file for the model to train on, providing significant data and twelve different features for the model to learn from. For preprocessing, data was ingested through a distributed framework. Data was then split into partitions, with 80% being used for training, 10% for validation, and 10% for testing. Once the data was prepared, it was sent to Missouri S&T's high-performance computing cluster, "The Mill," which provided the computational resources necessary to train the TCN model. Use of "The Mill" allowed efficient parallel processing of TCN operations to occur across multiple training epochs.

Our model also uses Temporal Convolutional Networks, allowing us to add external features, including temporal markers, weather variables and a holiday indicator, improving its contextual awareness when making predictions. The Temporal Convolutional Network architecture that we used also incorporates causal and dilated convolutions, residual connections, spatial dropout, and layer normalization. We also utilized sliding window segmentation in the model, allowing the effective dataset size to increase, while reducing overfitting of the model.

We use a few different types of criteria to determine the success of our model. Firstly, we compare the results of our model to related TCN models through the use of weighted average percentage error (WAPE), which was previously used to compare models in “Temporal Convolutional Networks Applied to Energy-Related Time Series Forecasting” [1]. There are also a couple of other metrics that we use in order to evaluate the success of our model including root mean squared error (RMSE) and mean absolute error (MAE). To decide if our model met the criteria for success, we wanted the model to achieve a WAPE of less than 10%, an RMSE of less than 3000, and an MAE of less than 2500. Since the final results of the model included a WAPE of 8.83%, an RMSE of 2,912.17, and a MAE of 2,252.37, we have met our criteria for success, and as a result have created a predictive model that can forecast long-term electricity consumption trends more accurately than other baseline models.

10 Conclusion

After reviewing related work in electricity forecasting and evaluating the available datasets, we developed and implemented a model for long-term monthly electricity consumption prediction. The primary focus of our work was to address a critical gap in the knowledge of this area due to the current focus being on short-term predictions of consumption rates. Through our model’s predictive capabilities we can better support more strategic energy planning helping manage electricity demand over the course of months/years.

TCNs were selected for our method due to their exhibited efficiency in modeling long-range dependencies in time-series data while processing data in parallel. These benefits are greatly appreciated when handling large-datasets. TCN’s ability to adequately take in exogenous

variables is also a benefit as it provides us with the ability to give more context to our data and its predictive output.

To improve the predictive accuracy we incorporated several methods to our approach. A sliding window segmentation enabled the model to learn from smaller periods of data aiding in reducing the potential for overfitting our model. Normalization shall aid in ensuring that external factors such as weather are weighed in a balanced manner along with mitigating outliers in our datasets.

Our final model achieved a strong performance based on our success thresholds. For example, our WAPE score of 8.83% indicates that our predictions deviate less than 9% on average from the actual consumptions when the model is tested. This informs us that our model is accurate, but there is always room for improvement. Capturing rapid transitions for electrical usage proved to have our model overestimate its predictions. Future work could be conducted on refining this scenario, with the current work serving as the foundation to be used. Rapid consumption changes are a likely event in real world scenarios.

This project successfully created a robust and scalable long-term electricity forecasting model utilizing TCNs. Our results verify that TCN-based approaches are valuable in tackling this issue and have value for energy planning initiatives. The work completed can serve as a foundation for future studies and offer insight into some of the challenges in Big-data analytics.

11 References

1. Lara-Benítez, P., Carranza-García, M., Luna-Romera, J. M., & Riquelme, J. C. (2020). Temporal convolutional networks applied to energy-related time series forecasting. *Applied Sciences*, 10(7), 2322. <https://doi.org/10.3390/app10072322>
2. Lee, [full author list]. (2022). A comparative study of forecasting electricity consumption using machine learning models. *Mathematics*, 10(8), 1329. <https://doi.org/10.3390/math1008132>
3. Matos, [full author list]. (2024). A machine learning-based electricity consumption forecast and management system for renewable energy communities. *Energies*, 17(3), 630. <https://doi.org/10.3390/en17030630>
4. Met Office. (n.d.). Historic station data. Retrieved December 10, 2025, from <https://www.metoffice.gov.uk/research/climate/maps-and-data/historic-station-data>
5. Shaikh, A. K., Nazir, A., Khalique, N., Shah, A. S., & Adhikari, N. (2023). A new approach to seasonal energy consumption forecasting using temporal convolutional networks. *Results in Engineering*, 19, 101296. <https://doi.org/10.1016/j.rineng.2023.101296>
6. Vidal Rod, A. (2024). Electricity consumption UK (2009–2024). <https://www.kaggle.com/datasets/albertovidalrod/electricity-consumption-uk-20092022>

7. Wang, Z., Hong, T., Li, H., & Piette, M. A. (2021). Predicting city-scale daily electricity consumption using data-driven models. *Advances in Applied Energy*, 2, 100025.
<https://doi.org/10.1016/j.adapen.2021.100025>